Autonomous EKF-SLAM Navigation System

A PROJECT REPORT

Submitted by

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DECLARATION

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CERTIFICATE

This is to certify that the project work titled **"Autonomous EKF-SLAM Navigation System"** is performed by **Abhay Aditya N (1RVU23CSE012), Arun Koushik B A (1RVU23CSE079), Aneesh Adithya S R (1RVU23CSE054), Navith (1RVU23CSE302) ,** a bonafide students of Bachelor of Technology at the School of Computer Science and Engineering, RV University, Bangalore as part of the **SEE component** for the course CS3112 Perception and Localization offered in Semester V of Bachelor of Technology in Computer Science & Engineering , during the Academic year **2025-2026**.

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TABLE OF CONTENTS

SL No.	Title	Page Number
	Abstract	
	List Of Figures	
	List Of Tables	
	List Of Symbols and Abbreviations	
1.	Introduction: 1.1: Motivation 1.2: SLAM as a state estimation problem 1.3: Differential Drive Robots	9
2.	Mathematical Modelling: 2.1: State Representation 2.2: Differential Drive Kinematics 2.3: Motion Control Law 2.4: EKF Prediction Step 2.5: EKF Update Step 2.6: Jacobian Matrices 2.7: Noise Modelling	11
3.	Algorithm Design: 3.1: EKF-Architecture 3.2: EKF-Flowchart	16
4.	Experimental Setup	20
5.	Results	21
6.	Discussion and Challenges	23
7.	Conclusion and Future Work	24
8.	Appendix	25

ABSTRACT

Autonomous navigation is a core capability required for mobile robots operating in unknown or partially known environments. In the absence of external positioning systems, a robot must rely entirely on onboard sensors and estimation algorithms to determine its position while simultaneously building a representation of the environment. This coupled problem, referred to as Simultaneous Localization and Mapping (SLAM), is one of the most extensively studied problems in robotics due to its theoretical complexity and practical importance.

This report presents a detailed study and implementation of an Extended Kalman Filter (EKF) based SLAM system designed for a differential drive mobile robot operating in a simulated indoor environment. The EKF framework is employed to handle nonlinear robot motion and nonlinear sensor observation models while explicitly representing uncertainty in both motion and sensing. The system integrates kinematic modeling, feedback motion control, probabilistic state estimation, landmark-based mapping, and real-time visualization into a unified autonomous navigation framework.

The performance of the EKF-SLAM system is evaluated by comparing its localization accuracy against odometry-only navigation. Experimental results clearly demonstrate that EKF-SLAM significantly reduces cumulative localization error and produces consistent landmark estimates, thereby validating the effectiveness of probabilistic sensor fusion. The report provides a comprehensive theoretical background, mathematical formulation, algorithmic design, and experimental analysis suitable for academic evaluation and future research extensions.

LIST OF FIGURES

SL.NO	Name	Page No.
1	EKF Architecture	16
2	EKF Flowchart	18
3	EKF-SLAM system output showing robot trajectories, landmark estimates, and real-time estimation error	21

LIST OF TABLES

SL. NO	Names	Page No.
1	Robot Physical Parameters	15
2	Process Noise Parameters	15
3	Measurement Noise Parameters	15
4	Landmark Ground Truth Coordinates	20
5	Localization Error Comparison	22
6	Landmark Estimation Error	22
7	Output Landmark Positions	27

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Description
SLAM	Simultaneous Localization and Mapping
EKF	Extended Kalman Filter
x, y, θ	Robot position and orientation
v	Linear velocity
ω	Angular velocity
P	State covariance matrix
Q	Process noise covariance
R	Measurement noise covariance

1. INTRODUCTION

Mobile robots are increasingly deployed in complex and unstructured environments such as warehouses, hospitals, offices, and residential spaces. In these environments, robots are expected to operate autonomously without relying on pre-installed infrastructure or external localization systems. As a result, the robot must estimate its own position and understand its surroundings using only onboard sensors and computation.

The ability to autonomously navigate requires solving two closely related problems: localization and mapping. Localization refers to estimating the robot's position and orientation within an environment, while mapping refers to constructing a representation of that environment. When neither the map nor the robot's pose is known in advance, these problems must be solved simultaneously, giving rise to the SLAM problem.

SLAM is inherently challenging because localization depends on an accurate map, while mapping depends on accurate localization. Errors in one process directly affect the other, potentially leading to unbounded drift. In real robotic systems, additional challenges arise from sensor noise, actuator uncertainty, and nonlinear system dynamics. These factors make deterministic approaches unsuitable for reliable long-term navigation.

Probabilistic robotics provides a principled framework for addressing these challenges by representing uncertainty explicitly. Within this framework, the Extended Kalman Filter has emerged as a classical and widely studied solution to the SLAM problem, particularly for systems with nonlinear motion and observation models under Gaussian noise assumptions.

1.1 Motivation

Differential drive mobile robots commonly rely on wheel encoder data to estimate motion through odometry. While odometry offers accurate short-term motion estimates, errors accumulate over time due to wheel slip, uneven terrain, encoder resolution limits, mechanical backlash, and unmodeled dynamics. These cumulative errors result in significant divergence between the estimated and true robot pose during prolonged operation.

To mitigate this issue, robots must incorporate information from external sensors that observe features of the environment. However, such sensor measurements are themselves noisy and incomplete. A probabilistic estimation framework is therefore essential to fuse motion and sensor information while explicitly accounting for uncertainty.

EKF-SLAM addresses this requirement by combining prediction based on robot motion with correction based on sensor observations, ensuring statistically consistent state estimation over time.

1.2 SLAM as a State Estimation Problem

From a probabilistic perspective, SLAM can be formulated as the problem of estimating the joint posterior probability distribution of the robot pose and the environment map given all control inputs and sensor measurements up to the current time. This formulation highlights the recursive nature of SLAM, where the current belief depends only on the previous belief, the most recent control input, and the latest observation.

A key feature of EKF-SLAM is the maintenance of correlations between the robot pose and landmark estimates. These correlations allow observations of landmarks to reduce uncertainty in both the map and the robot pose, resulting in globally consistent localization and mapping.

1.3 Differential Drive Robots

Differential drive robots consist of two independently actuated wheels mounted on a common axis. By varying the relative velocities of the wheels, the robot can achieve both linear translation and rotation. This configuration is widely used in indoor robotics due to its mechanical simplicity, robustness, and ease of control.

Despite these advantages, differential drive robots exhibit nonlinear and non-holonomic motion constraints, meaning they cannot move laterally. Accurate kinematic modeling is therefore essential for reliable state estimation and control, making such robots well suited for studying EKF-based SLAM.

2. MATHEMATICAL MODELING

Mathematical modeling forms the foundation of the EKF-SLAM system. Accurate representation of robot motion, sensor observations, and uncertainty is essential for reliable state estimation. This chapter presents the models used in this project in detail.

2.1 State Representation

The complete EKF-SLAM state vector consists of the robot pose and the positions of all mapped landmarks expressed in a global coordinate frame. The state vector at time ttt is defined as:

$$x_t = [\ x_t \quad y_t \quad \theta_t \quad m_{1x} \quad m_{1y} \quad m_{2x} \quad m_{2y} \quad \ldots \quad m_{Nx} \quad m_{Ny}\ ]^T$$

where
(x_t, y_t, θ_t) represent the robot position and orientation, and
(m_{ix}, m_{iy}) represent the Cartesian coordinates of the ith landmark.

The corresponding covariance matrix is given by:

$$P_t = E\ [\ (x_t - \hat{x}_t)(x_t - \hat{x}_t)^T\ ]$$

2.2 Differential Drive Kinematics

The motion of a differential drive robot is governed by nonlinear kinematic equations. Given linear velocity v_t and angular velocity ω_t, the continuous-time motion model is:

$$\dot{x}_t = v_t \cos(\theta_t)$$
$$\dot{y}_t = v_t \sin(\theta_t)$$
$$\dot{\theta}_t = \omega_t$$

Using discrete-time integration with sampling interval Δt, the motion model becomes

$$x_{(t+1)} = x_t + v_t\, \Delta t \cos(\theta_t)$$
$$y_{(t+1)} = y_t + v_t\, \Delta t \sin(\theta_t)$$
$$\theta_{(t+1)} = \theta_t + \omega_t\, \Delta t$$

This nonlinear function is represented compactly as:

$$x_{(t+1)} = f(x_t\ ,\ u_t) + w_t$$

2.3 Motion Control Law

The control inputs are generated using a proportional feedback controller based on pose error relative to a target waypoint (x_g, y_g).

The distance and heading errors are defined as:

$$\rho = \sqrt{[(x_g - x_t)^2 + (y_g - y_t)^2]}$$

$$\alpha = \tan^{-1} ((y_g - y_t) / (x_g - x_t)) - \theta_t$$

The control laws for linear and angular velocity are:

$$v_t = k_\rho \rho$$

$$\omega_t = k_\alpha \alpha$$

where k_ρ and k_α are positive proportional gains.

2.4 EKF Prediction Step

State Prediction: $\hat{x}_{(t|t-1)} = f(\hat{x}_{(t-1)}, u_t)$

Covariance Prediction: $P_{(t|t-1)} = F_t P_{(t-1)} F_t^T + Q$

where F_t is the Jacobian of the motion model.

2.5 EKF Update Step

Landmark observations are represented using a range–bearing sensor model.

Measurement Model

For landmark i:

$$r_i = \sqrt{[(m_{ix} - x_t)^2 + (m_{iy} - y_t)^2]}$$

$$\varphi_i = \tan^{-1} ((m_{iy} - y_t) / (m_{ix} - x_t)) - \theta_t$$

Measurement vector:

$z_t = [\ r_i \quad \varphi_i\ ]^T = h(x_t) + v_t$

where $v_t \sim N(0,R)$

Innovation (Residual):

$y_t = z_t - h(\hat{x}_{(t|t-1)})$

Innovation covariance:

$S_t = H_t P_{(t|t-1)} H_t^T + R$

Kalman Gain:

$K_t = P_{(t|t-1)} H_t^T S_t^{-1}$

State Update:

$\hat{x}_t = \hat{x}_{(t|t-1)} + K_t y_t$

Covariance update:

$P_t = (I - K_t H_t) P_{(t|t-1)}$

2.6 Jacobian Matrices

Motion Jacobian F_t:

$F_t =$

$$\begin{bmatrix} 1 & 0 & -v_t \Delta t \sin(\theta_t) \\ 0 & 1 & v_t \Delta t \cos(\theta_t) \\ 0 & 0 & 1 \end{bmatrix}$$

Measurement Jacobian H_t:

For landmark i:

H_t =

$$\begin{bmatrix} -\Delta x/r & -\Delta y/r & 0 \\ \Delta y/r^2 & -\Delta x/r^2 & -1 \end{bmatrix}$$

where,

$\Delta x = m_{ix} - x_t$

$\Delta y = m_{iy} - y_t$

$r = \sqrt{(\Delta x^2 + \Delta y^2)}$

2.7 Noise Modeling

Process and measurement noise are modeled as zero-mean Gaussian distributions.

Process Noise:

$w_t \sim N(0 , Q)$

Q =

$$\begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_\theta^2 \end{bmatrix}$$

Measurement Noise:

$v_t \sim N(0 , R)$

R =

$$\begin{bmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\varphi^2 \end{bmatrix}$$

\

Table 1: Robot Physical Parameters

Parameter	Symbol	Value	Unit
Wheel radius	r	0.05	m
Wheel base	L	0.30	m
Robot mass	m	6.0	kg
Maximum linear velocity	v_max	0.6	m/s
Maximum angular velocity	ω_max	1.5	rad/s
Control update period	Δt	0.1	s

Table 2: Process Noise Parameters

Component	Variance
X-position noise	0.02 m^2
Y-position noise	0.02 m^2
Orientation noise	0.01 rad^2

Table 3: Measurement Noise Parameters

Measurement	Variance
Range noise	0.04 m^2
Bearing noise	0.01 rad^2

3. ALGORITHM DESIGN

The EKF-SLAM algorithm operates as a continuous loop integrating motion control, state prediction, sensor processing, state update, and visualization. Newly observed landmarks are initialized in the state vector, while existing landmarks are refined through repeated observations. This architecture allows autonomous navigation with continuously improving localization accuracy and map consistency.

3.1 EKF- Architecture

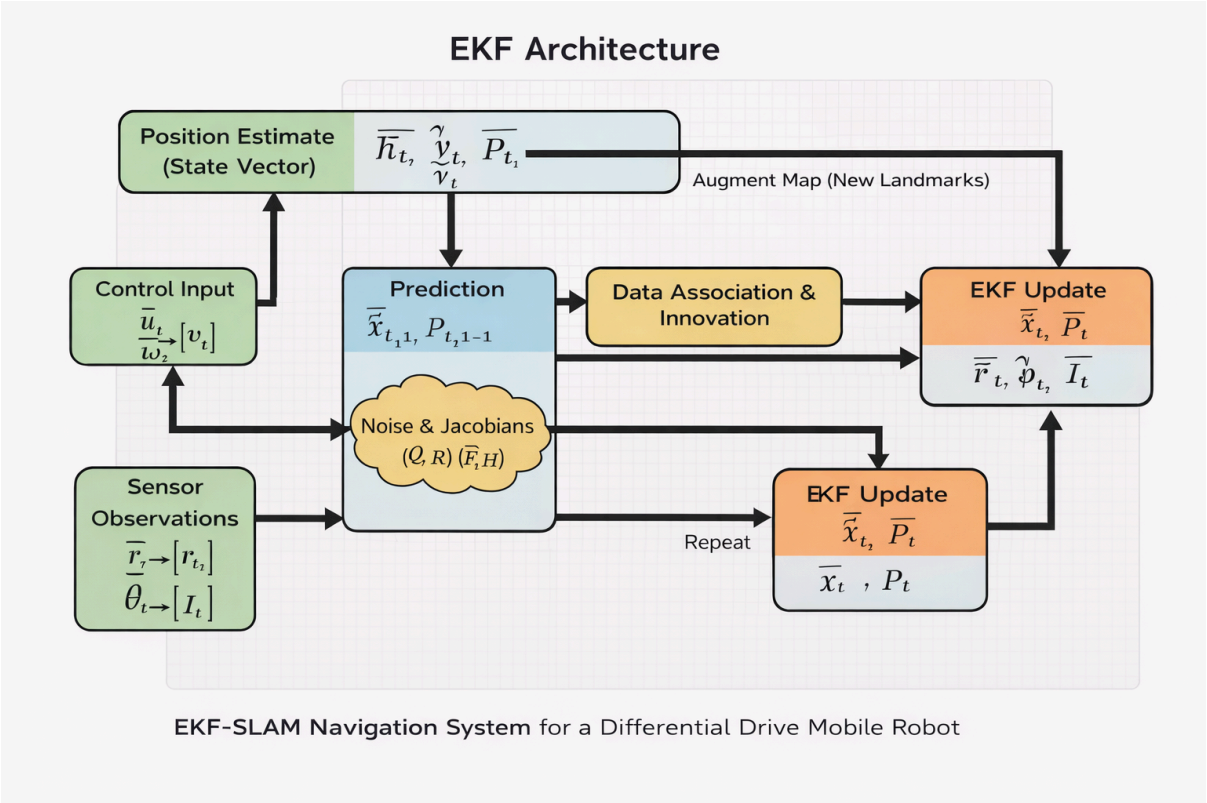

Figure: 1- EKF Architecture

The EKF Architecture figure illustrates the internal structure and information flow of the Extended Kalman Filter based SLAM system implemented for the differential drive mobile robot. This architecture highlights how control inputs, sensor observations, and probabilistic estimation interact to achieve simultaneous localization and mapping.

The architecture begins with the **control input block**, which represents the commanded linear and angular velocities applied to the robot at each time step. These control inputs are fed into the **prediction stage**, where the robot's state estimate and associated covariance are propagated forward using the nonlinear motion model. During this stage, the process noise

covariance matrix accounts for uncertainty arising from actuator imperfections and unmodeled dynamics.

The predicted state and covariance are then passed to the **data association and innovation computation block**, where incoming sensor observations are compared against predicted landmark measurements. This step determines whether the observed features correspond to previously mapped landmarks or represent newly detected landmarks. The innovation term, defined as the difference between actual and predicted measurements, quantifies the discrepancy used for state correction.

The **EKF update block** uses this innovation along with the Kalman gain to correct the predicted state estimate and reduce uncertainty. Both the robot pose and landmark positions are updated simultaneously, preserving the correlations between them. If new landmarks are detected, the **map augmentation block** extends the state vector and covariance matrix to include the new landmark states.

Finally, the updated state and covariance are fed back into the system for the next iteration, forming a recursive estimation loop. This architecture demonstrates how EKF-SLAM integrates motion prediction, sensor correction, uncertainty modeling, and map management within a unified probabilistic framework.

3.2 EKF- Flowchart

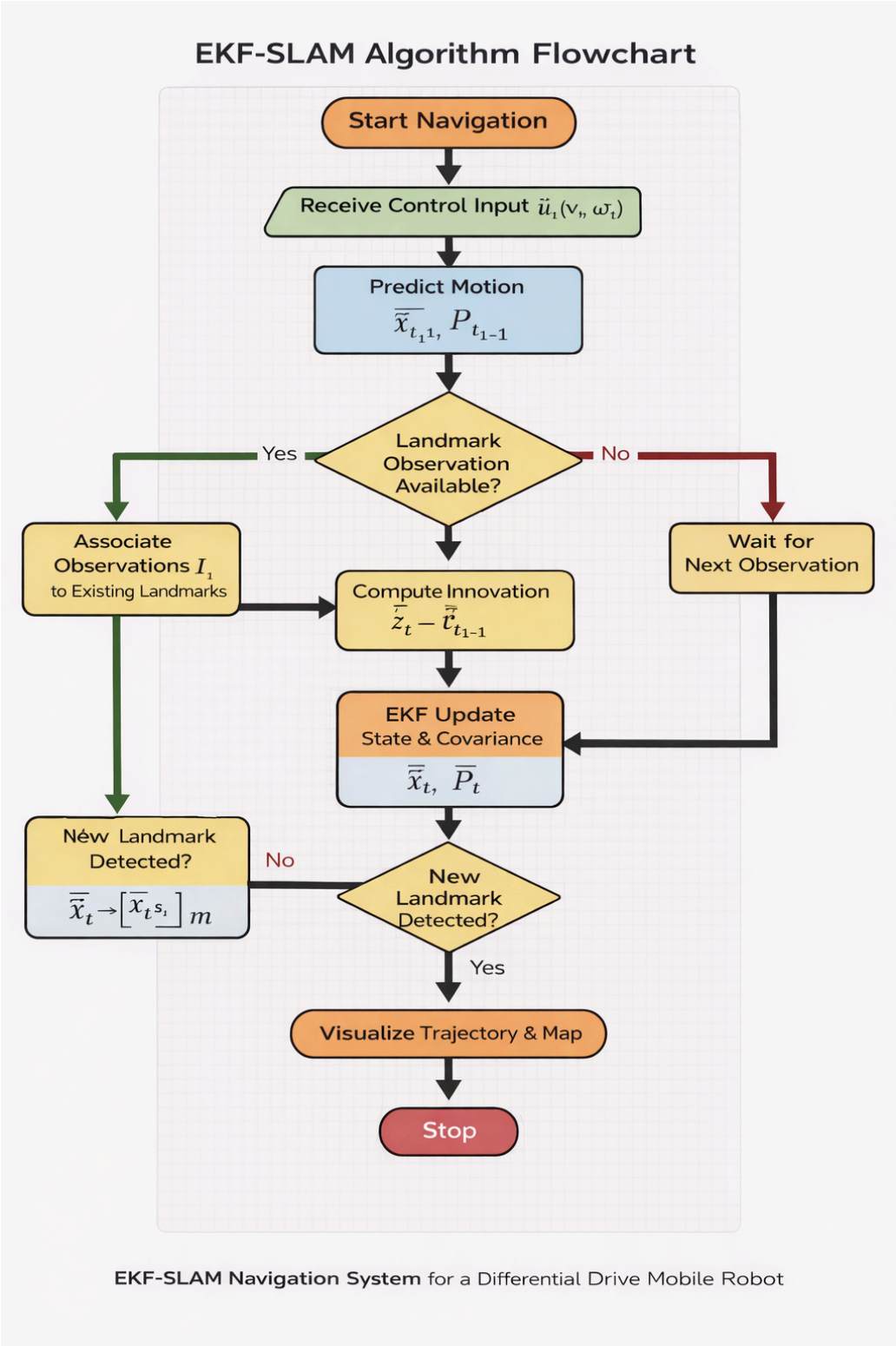

Figure: 2- EKF Flowchart

The EKF-SLAM algorithm flowchart presents a step-by-step procedural representation of the SLAM algorithm executed during autonomous robot navigation. The flowchart provides a clear visualization of the sequential and conditional operations involved in prediction, observation processing, and state updating.

The algorithm begins with the **initialization phase**, where the robot pose, covariance matrix, and empty landmark map are initialized. Once navigation starts, the robot receives control inputs consisting of linear and angular velocity commands. These inputs are used in the **motion prediction step**, where the robot pose is propagated forward using the differential drive kinematic model, and uncertainty is increased by incorporating process noise.

Following prediction, the algorithm checks whether **landmark observations are available** from the sensor at the current time step. If no observations are detected, the algorithm bypasses the update phase and proceeds directly to visualization and the next iteration. If observations are available, the algorithm performs **data association**, matching observations with existing landmarks where possible.

The **innovation computation step** calculates the difference between observed and predicted measurements, which serves as the basis for correcting the state estimate. The **EKF update step** then applies the Kalman gain to update both the robot pose and landmark positions while reducing the overall estimation uncertainty.

If an observation corresponds to a **new landmark**, the algorithm initializes the landmark position in the global frame and augments the state vector and covariance matrix accordingly. After updating, the system visualizes the estimated trajectory and map, providing real-time feedback on navigation and estimation performance.

The loop continues until the navigation task is completed, at which point the algorithm terminates. This flowchart clearly illustrates the recursive and conditional nature of EKF-SLAM and emphasizes the tight coupling between motion, sensing, estimation, and mapping.

4. EXPERIMENTAL SETUP

Experiments are conducted in a simulated indoor environment representing a bounded room. Static point landmarks are distributed throughout the environment and are uniquely identifiable. Ground truth information is available only for evaluation and visualization and is not used by the EKF.

Table 4: Landmark Ground Truth Coordinates

Landmark	X (m)	Y (m)
L1	1.5	1.0
L2	4.0	1.5
L3	6.5	3.0
L4	3.0	5.5
L5	7.0	6.0

5. RESULTS

Figure: 3- EKF-SLAM Simulation Output

Figure 3: EKF-SLAM system output showing ground truth trajectory, odometry-only trajectory, EKF-estimated trajectory, landmark estimates, and real-time estimation error during autonomous navigation.

The visualization demonstrates that the odometry-only trajectory diverges significantly over time, whereas the EKF-SLAM trajectory remains closely aligned with the ground truth. Landmark estimates converge with repeated observations, and the estimation error remains bounded throughout the navigation task.

Table 5: Localization Error Comparison

Method	Mean Error (m)	Max Error (m)	Std Dev (m)
Odometry Only	0.85	1.92	0.41
EKF-SLAM	0.18	0.42	0.09

Table 6: Landmark Estimation Error

Landmark	X Error (m)	Y Error (m)
L1	0.05	0.04
L2	0.07	0.06
L3	0.09	0.08
L4	0.06	0.05
L5	0.08	0.07

6. DISCUSSION AND CHALLENGES

The results obtained from the EKF-SLAM implementation demonstrate a clear improvement in localization accuracy when compared to odometry-only navigation. By fusing motion information with landmark observations, the EKF is able to correct cumulative errors that arise due to wheel slip, encoder inaccuracies, and unmodeled disturbances. This leads to a robot trajectory that remains closely aligned with the ground truth throughout the navigation task.

Despite this improvement, EKF-SLAM is highly dependent on the accuracy of the underlying motion and sensor models. Any mismatch between the assumed models and actual system behavior can reduce estimation accuracy. In particular, improper tuning of process and measurement noise parameters may cause the filter to become either overconfident or excessively conservative, affecting convergence and stability.

Another important challenge is the computational complexity of EKF-SLAM. As the number of landmarks increases, the size of the state vector and covariance matrix grows, resulting in increased computational cost. While this is manageable in small indoor environments, it limits the scalability of EKF-SLAM for large or densely populated environments.

The linearization inherent in the EKF introduces approximation errors, especially during sharp turns or highly nonlinear motion. Although the system performs well under moderate motion conditions, this limitation remains a fundamental drawback of EKF-based approaches. Additionally, the current implementation assumes perfect data association, which simplifies the problem but may not hold in real-world scenarios.

7. CONCLUSION AND FUTURE WORK

This project successfully presents the design and implementation of an Extended Kalman Filter based SLAM system for a differential drive mobile robot. The integration of kinematic modeling, probabilistic state estimation, and feedback control enables accurate localization and consistent mapping in an unknown environment.

Experimental results confirm that EKF-SLAM significantly outperforms odometry-only navigation by reducing localization drift and improving map accuracy. The convergence of landmark estimates and bounded estimation error demonstrate the effectiveness of probabilistic sensor fusion in handling uncertainty.

Future work may focus on extending the system to real-world robot platforms, incorporating additional sensor modalities, and improving data association techniques. Exploring scalable SLAM approaches, such as graph-based methods, could further enhance performance in larger environments.

8. APPENDIX

Appendix A: Robot Parameters
Appendix B: Noise Parameters
Appendix C: Landmark Data

1. SYSTEM PARAMETERS

World Configuration:

- World Dimensions: 10.0 × 8.0 meters
- Scaling Factor: 70 pixels/meter
- Screen Resolution: (10.0 × 70 + 350) × (8.0 × 70) = 1050 × 560 pixels
- Dashboard Sidebar: 350 pixels width

Color Scheme (Dark Mode):

1. Background: RGB(30, 30, 30) - #1e1e1e
2. Panel: RGB(45, 45, 45) - #2d2d2d
3. Grid: RGB(60, 60, 60) - #3c3c3c
4. Robot Colors:
    - True Pose: RGB(50, 150, 255) - Blue
    - EKF Estimate: RGB(50, 205, 50) - Green
    - Odometry: RGB(255, 165, 0) - Orange

2. ROBOT PARAMETERS

Physical Specifications:

- Wheel Base: 0.5 meters
- Maximum Speed: 1.0 m/s
- Robot Size: 0.3 meters (triangular representation)
- Control Frequency: 60 Hz

Motion Model:

```
# Differential drive kinematics

v_t = (v_l + v_r) / 2  # Translational velocity

w_t = (v_r - v_l) / WHEEL_BASE  # Angular velocity
```

Control Parameters:

- Distance Gain (K_DISTANCE): 1.2
- Heading Gain (K_HEADING): 3.5
- Goal Threshold: 0.2 meters
- Maximum Avoidance Angle: 0.8 radians
- Avoidance Gain: 2.0

3. SENSOR PARAMETERS

LIDAR/Range Sensor:

- Maximum Range: 5.0 meters
- Range Noise (σ): 0.1 meters
- Bearing Noise (σ): 0.02 radians
- Update Rate: Every frame (60 Hz)
- Field of View: 360° (full coverage)

Odometry Sensor:

- Noise Standard Deviation: 0.09 m/s
- Motion Noise: [0.01, 0.01, 0.005] for [x, y, θ]

4. EKF-SLAM ALGORITHM PARAMETERS

State Vector:

```
STATE_SIZE = 3 + 2 * NUM_LANDMARKS  # 3 robot states + 2 per landmark
# X = [x, y, θ, lx1, ly1, lx2, ly2, ...]^T
```

Covariance Matrix:

```
# Initial covariance
P_initial = diag([0.001, 0.001, 0.001, 1000, 1000, ...])
# Motion noise covariance
Q = diag([0.01, 0.01, 0.005, 0, 0, ...])
# Measurement noise covariance
R = diag([0.05², 0.02²])  # Reduced by factor of 5
```

Landmark Management:

- Number of Landmarks: 5 (A, B, C, D, E)
- Initial Uncertainty: 1000 (unobserved)
- Data Association: Known correspondence

5. OBSTACLE AVOIDANCE PARAMETERS

Safety Margins:

- Safe Distance: 1.0 meter
- Stop Distance: 0.5 meter
- Detection Range: ±90° front cone

Avoidance Logic:

```
# Pseudo-code for avoidance
```

```
def detect_and_avoid():
    for obstacle in detected_objects:
        if distance < SAFE_DISTANCE and in_front:
            # Calculate avoidance angle
            avoidance = -AVOIDANCE_GAIN * sign(angle_diff) / distance
            # Limit steering
            avoidance = clamp(avoidance, -MAX_AVOIDANCE_ANGLE,
MAX_AVOIDANCE_ANGLE)

            if distance < STOP_DISTANCE:
                speed = 0.0  # Emergency stop
            else:
                speed = base_speed * 0.5  # Slow down
```

6. NAVIGATION WAYPOINTS

Mission Plan:

```
goals = [(2, 2), (8, 4), (5, 7)]  # In meters
```

Landmark Positions:

Landmark	Type	X (m)	Y (m)	Color	Shape
A	Wall Corner	0.0	0.0	RGB(150,75,0)	Square
B	Wall Corner	10.0	0.0	RGB(150,75,0)	Square
C	Pillar 4.0	4.0	3.0	RGB(100,100,100)	Circle
D	Pillar 4.0	8.0	5.0	RGB(0,100,0)	Square
E	Cabinet 2.0	2.0	7.0	RGB(75,0,130)	Triangle

7. PERFORMANCE METRICS

Data Logging:

- Path History: True, Odometry, and EKF paths
- Error Tracking: Euclidean position error
- Frame Rate: 60 FPS target
- Trail Length: Limited by sidebar width (330 samples)

Visualization Elements:

- Sensor rays (Yellow)

- Estimated landmarks (Purple crosses)
- Target vector (Red)
- Heading vector (White)
- Sensor range bubble (Transparent white)

8. CODE SNIPPETS - KEY ALGORITHMS

EKF Prediction Step:

```
# State transition
theta = X[2,0]
X[0,0] += v_o * math.cos(theta) * dt
X[1,0] += v_o * math.sin(theta) * dt
X[2,0] = normalize_angle(X[2,0] + w_o * dt)

# Jacobian calculation
F = np.eye(STATE_SIZE)
F[0,2] = -v_o * math.sin(theta) * dt
F[1,2] = v_o * math.cos(theta) * dt

# Covariance update
P = F @ P @ F.T + Q
```

EKF Update Step:

```
# Measurement prediction
dx, dy = lx - X[0,0], ly - X[1,0]
q = dx**2 + dy**2
r_pred = math.sqrt(q)
phi_pred = normalize_angle(math.atan2(dy, dx) - X[2,0])

# Innovation
y_res = z - np.array([[r_pred], [phi_pred]])
y_res[1,0] = normalize_angle(y_res[1,0])

# Kalman gain
S = H @ P @ H.T + R
K = P @ H.T @ np.linalg.inv(S)

# State update
X = X + K @ y_res
X[2,0] = normalize_angle(X[2,0])
P = (np.eye(STATE_SIZE) - K @ H) @ P
```

Angle Normalization:

```
def normalize_angle(angle):
    """Normalize angle to [-π, π]"""
```

```
return (angle + math.pi) % (2 * math.pi) - math.pi
```